Temperature field prediction in additive manufacturing process

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1 Introduction

Additive Manufacturing (AM), also known as 3D printing, is a disruptive manufacturing technology that has grown rapidly in the manufacturing industry and has gained a lot of attention owing to its ability to manufacture parts with complex features by using a layer-by-layer approach. This approach offers an extraordinary freedom of design while offering the potential of (nearly) zero-waste manufacturing solution. AM serves the industries in various domains, such as aerospace, defense, medicine, automotive, tooling and jewellery. The variability in the final product quality is however one of the major hurdles to the widespread application of such techniques in production environment. In particular, the mechanical properties and the quality of the manufactured part largely depend on the distribution of temperature fields during the AM process. Numerical simulations, such as finite element analyses, are commonly used to simulate the thermal history during the AM process, but they are known to be expensive and time-consuming (taking up to several hours or even days for a single simulation) and hence they cannot be used in real-time. In this context, we propose a machine learning (ML) approach for the fast and accurate prediction of the thermal field evolution. Our approach consists of a variational autoencoder (VAE) that encodes the thermal fields into a latent space, combined with a recurrent neural network (RNN) that simulates the temporal process in the latent space. Our VAE-RNN implementation is available at `https://github.com/VicD1999/AM_process`.

2 Problem statement

We assume a simplified AM setting where the manufactured part is a 2D rectangular metallic piece of fixed size. This part is manufactured through a powder bed fusion process, during which a laser beam moves on the top of the part at a constant speed and adds layers of material by melting the deposited powder. The laser performs several passes, with potentially a break after each pass where the laser is turned off. The main parameters defining the execution of the process are: (1) the laser nominal power and (2) the break time. Both parameters remain constant during the whole process. The objective is to provide an ML model able to predict the temporal evolution of the temperature field (Figure 1), for any given set of process parameters (i.e., laser power and break time).

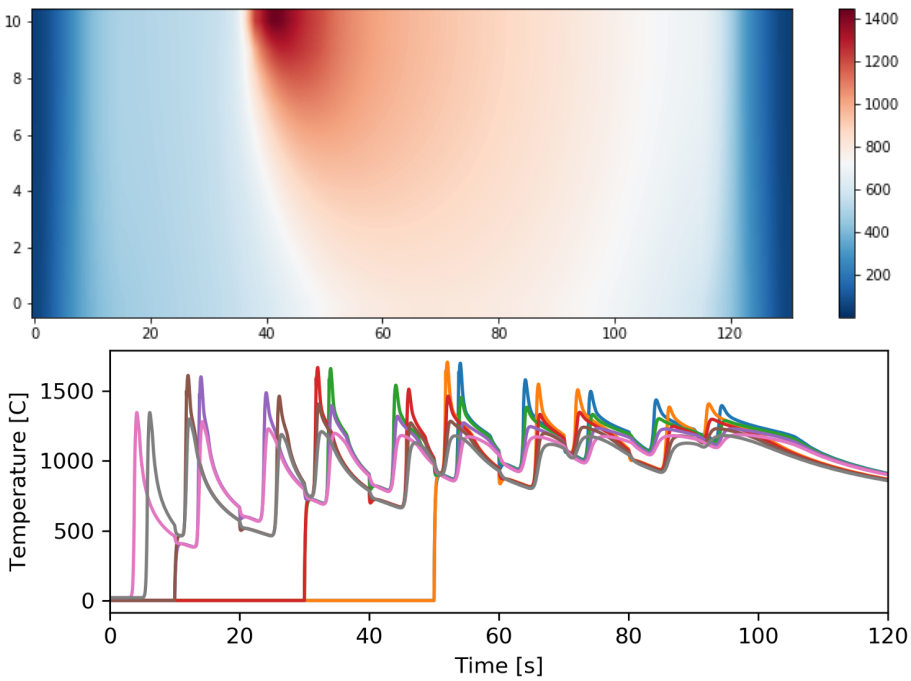

Fig. 1. The top figure shows the 2D thermal field at one time point. The bottom figure shows the temporal evolution of the temperature for 8 thermocouples (one color per thermocouple), which are located at different layers in the manufactured part. The temperature is initialized to 20 degrees and starts to evolve when the corresponding layer is added. Temperature peaks can be observed at the passage of the laser.

3 Methods

3.1 Simulation data

In order to generate the datasets for training and testing the ML models, 256 finite element-based thermal simulations are performed using Cenaero's virtual manufacturing software, Morfeo. For each simulation, the laser power is sampled from the interval $[50 - 250W]$ and the break time is sampled from the interval $[0 - 10s]$. The part initially consists of a support layer. The laser then moves at a constant speed along the top of the part, with a break after each pass, and a new layer of material is added after each pass and break. Ten layers are added with this setting before the laser returns to its initial position, followed by a cooling phase. In each simulation, the target data consists of the simulated temperature, at each time point, at 131 equally spaced points in each layer. At each time point t, we format the target data as a 2D grid $X_t \in \mathbb{R}^{11\times131}$ (Figure 1, top) and we use binary masks to identify grid points in layers that are not yet constructed. We use a 80%-10%-10% split to divide the 256 simulations into respectively training, validation and test sets. Temperature grids are standardized before training so that the temperature at each grid point has zero mean and unit variance.

3.2 ML-based approach for temperature field prediction

Inspired by the World Model architecture [1], our proposed approach consists of two components:

1. A variational autoencoder (VAE) [2] that projects temperature grids into a latent space. More specifically, the encoder part of the VAE maps a given temperature grid $X \in \mathbb{R}^{11\times131}$ into a normal distribution $q(\mathbf{z}|X;\psi) = \mathcal{N}(\mathbf{z};\boldsymbol{\mu},\boldsymbol{\sigma}^2 I)$ over the latent space, where $\mathbf{z} \in \mathbb{R}^H$ is the latent vector, and $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}^2$ are the outputs of a neural network $\mathrm{NN}_\psi : \mathbb{R}^{11\times131} \to \mathbb{R}^H \times \mathbb{R}^H$ with parameters ψ. The decoder part of the VAE consists of a second neural network $\mathrm{NN}_\theta : \mathbb{R}^H \to \mathbb{R}^{11\times131}$ with parameters θ, that reconstructs the temperature grid from a given latent vector. In our experiments, we set the latent dimension H to 64.
2. A recurrent neural network (RNN) that predicts the future latent vector $\mathbf{z}_{t+1}$, given the current latent vector $\mathbf{z}_t$ and some current laser-related features $\mathbf{f}_t$. Note that in addition to the latent vector $\mathbf{z}_t$, the RNN also propagates through time a hidden state $\mathbf{h}_t$, that we initialize with 0's. More formally, the RNN is a parametrized function $\mathrm{RNN}_\phi : \mathbb{R}^H \times \mathbb{R}^F \to \mathbb{R}^H$, where F is the number of laser-related features, and we have $\mathbf{z}_{t+1} = \mathrm{RNN}_\phi(\mathbf{z}_t, \mathbf{f}_t)$, where we have left out the hidden state $\mathbf{h}$ for notation clarity. In our experiments, we use $F = 3$ laser-related features, which are the laser current position in the horizontal axis (x-coordinate), the laser current position in the vertical axis (y-coordinate) and the current laser power.

Hence, while the VAE models the process in space, the RNN models the temporal process in the latent space. Given a trained VAE and a trained RNN, the following procedure is used to simulate the evolution of the temperature grid: the initial grid X_1 is set to 20 degrees at all the points and then encoded into an initial latent vector $\mathbf{z}_1 = \mathrm{NN}_\psi(X_1)$. The RNN is then used to predict the future latent vectors $\mathbf{z}_2, \ldots, \mathbf{z}_T$, with $\mathbf{z}_t = \mathrm{RNN}_\phi(\mathbf{z}_{t-1}, \mathbf{f}_{t-1}), t = 2, \ldots, T$, and each of them is transformed back into a temperature grid $X_t = \mathrm{NN}_\theta(\mathbf{z}_t)$.

Architecture

- VAE encoder NN_ψ:
  - 3 convolutional layers with respectively 16, 32 and 64 channels, each with a stride of 2 and a kernel size of 3, and each followed by batch normalization and leaky ReLU activation;
  - 2 fully connected layers of 128 units with leaky ReLU activations;
  - 2 parallel linear layers that respectively output the mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\sigma}^2$ of the latent normal distribution q.
- VAE decoder NN_θ:
  - 2 fully connected layers of respectively 128 and 2176 units with leaky ReLU activations;
  - 3 transposed convolutional layers with respectively 64, 32 and 16 channels, each with a stride of 2 and a kernel size of 3, and each followed by batch normalization and leaky ReLU activation;
  - 1 convolutional layer with 1 channel and a kernel size of 3, that outputs the temperature grid.

– RNN:
  - 1 GRU recurrent layer with a hidden state of size 1024, which takes as inputs the current latent vector, the current hidden state, and the current laser coordinates and power;
  - 2 fully connected layers of respectively 512 and 64 units with leaky ReLU activations.

Training While the VAE and the RNN work together, they are trained *separately* from each other. Training the whole model on sequences of grids would indeed be very long, while training only the VAE on all the grids can benefit from the GPUs which fasten the training, and the RNN is hence trained on sequences where the sample dimension is much smaller (64 vs 11*131).

First, the VAE is trained using the following regularized loss function:

$$(\hat{\psi}, \hat{\theta}) = \arg\min_{(\psi,\theta)} \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}_{\mathbf{z}_t \sim q(\mathbf{z}_t|X_t;\psi)} \left[||X_t - \mathrm{NN}_\theta(\mathbf{z}_t)||_2^2 \right] + \lambda \mathrm{KL} \left[q(\mathbf{z}_t|X_t;\psi) || \mathcal{N}(\mathbf{z}_t; 0, I) \right], \quad (1)$$

where λ is a regularization coefficient. The first term in this loss function is the grid reconstruction error and the second term is the Kullback–Leibler divergence between the latent distribution and the standard normal distribution.

Once the VAE is trained, the RNN is trained by minimizing the MSE loss on the decoded grids:

$$\hat{\phi} = \arg\min_{\phi} \frac{1}{T} \sum_{t=1}^{T} ||X_t - \mathrm{NN}_{\hat{\theta}}(\mathrm{RNN}_\phi(\mathbf{z}_{t-1}, \mathbf{f}_{t-1}))||_2^2, \quad (2)$$

where the initial latent vector is set as $\mathbf{z}_1 = \mathrm{NN}_{\hat{\psi}}(X_1)$, with all the points in the grid X_1 set to 20 degrees.

The training of both the VAE and RNN are performed using the Adam optimizer and learning rate cosine annealing.

4 Results

The predictions of the VAE-RNN model are shown, for one simulation, in Figure 2 (for the whole grid at one time point) and in Figure 3 (for two grid points across time). One can observe that the highest (relative) errors occur at the borders of the part, where the laser does not pass. This is due to the fact that our model currently does not differentiate between points corresponding to melted powder (and hence becoming metallic material) and points corresponding to unmelted powder. High errors are also observed around the position of the laser, corresponding to temperature peaks that are notoriously hard to predict.

On the test set (comprising 25 simulations), the mean squared error is 369.93 (corresponding to an average absolute error of 9.32 degrees) and the mean relative error is 2.24%.

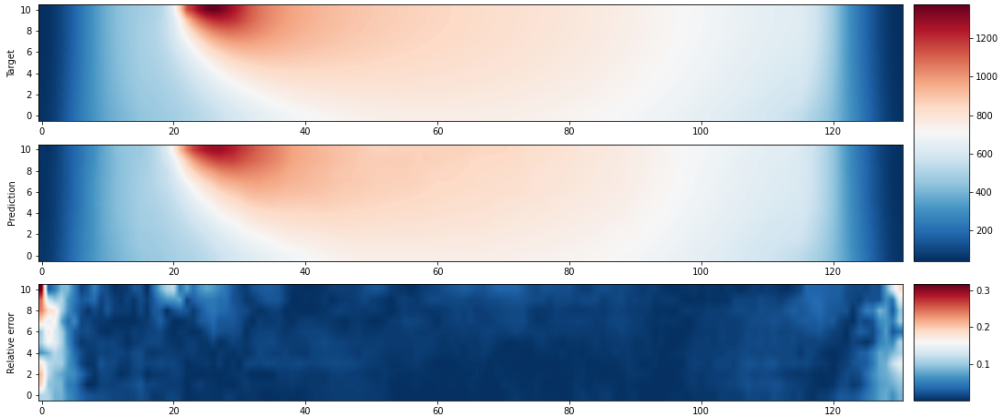

Fig. 2. This figures shows, at one time point in one simulation, the true temperature grid (top), the temperature predicted by our VAE-RNN model (middle) and the relative difference between them (bottom). The highest errors appear at the borders of the manufactured part (where there is no melting) and around the position of the laser.

5 Conclusion

We propose a data-driven approach for temperature prediction during the AM process, in which a VAE (modelling the process in space) is combined with a RNN (modelling the process in time inside a reduced latent space). This approach shows promising results in a simplified AM setting.

Future works include the extension of our approach to more complex settings including manufactured parts with variable geometries and sizes, as well as 3D parts.

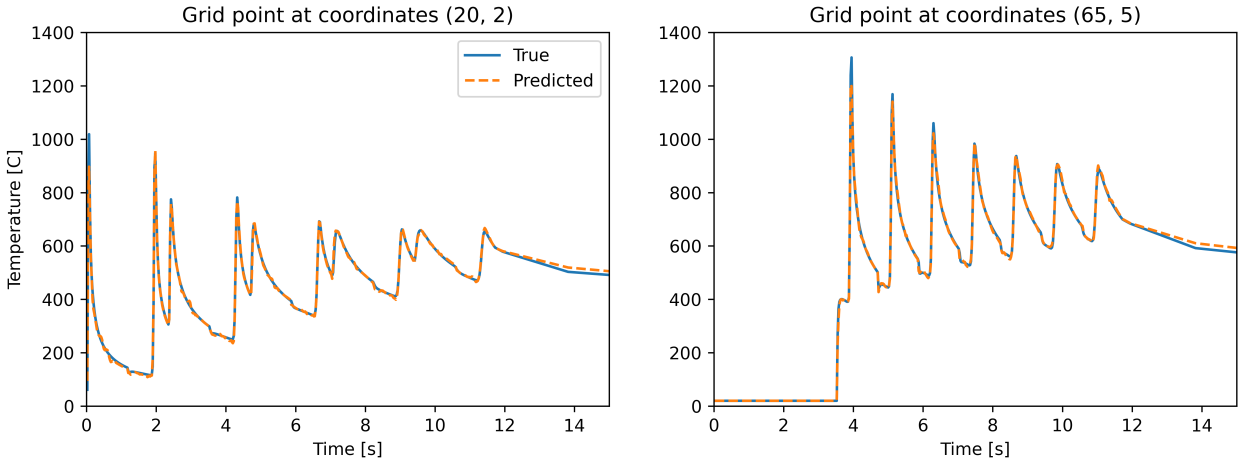

Fig. 3. Time evolution of predicted versus true temperatures, for two grid points in one simulation.

References

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